From World Models to Governed World-Forming Intelligence - A Safety-Oriented Architecture for Persistent, Self-Revising AGI

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WORLD Constitution, Residual Memory, Revision Depth, and Constitutional Governance

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Abstract

Artificial intelligence is increasingly acquiring capabilities that were once studied separately: generative reasoning, tool use, planning, persistent memory, self-evaluation, world modeling, mechanistic measurement, and increasingly autonomous operation. Yet most current AI still reasons largely inside representational worlds whose task boundaries, variables, interfaces, and success conditions are supplied by humans. A qualitatively different problem appears when an intelligent system begins to maintain and revise the effective WORLD within which its own reasoning occurs.

This article develops a safety-oriented architecture for that transition. It combines two preceding frameworks. The first represents an Effective WORLD as

$$\mathcal{W} = (V,F,C,M,O), \quad (0.1)$$

where V denotes effective distinctions or abstractions, F operational dynamics and consequence, C compositional coherence, M measurable realization, and O the perspective of an embedded bounded observer. The second describes runtime control through five regimes,

$$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G, \quad (0.2)$$

corresponding to Generation, Activation, Closure, Selection, and Retention, together with dual historical ledgers for admitted trace L^+ and unresolved residual L^-.

The combination suggests a stronger capability than conventional world modeling: **world-forming intelligence**, defined here as the capacity to construct, inhabit, maintain, criticize, and selectively revise the Effective WORLD within which an agent reasons and acts.

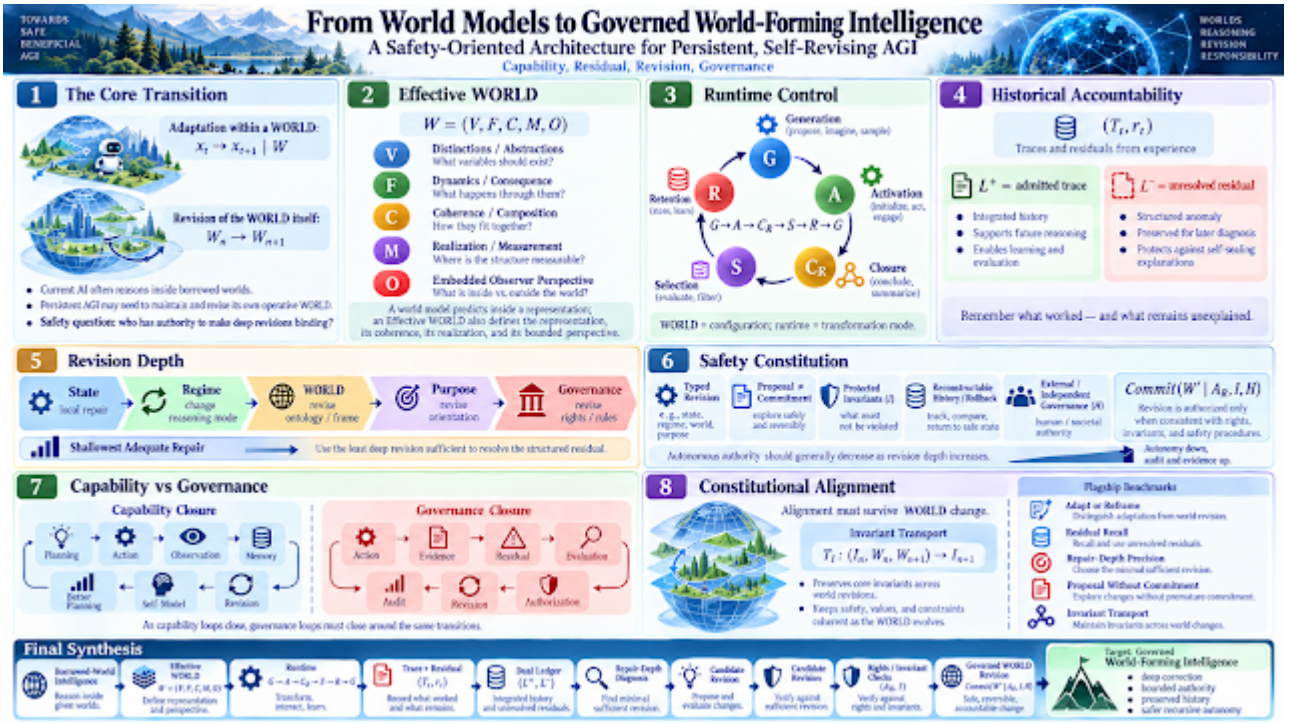
Such capability is intrinsically dual-use. It may allow future AGI to escape obsolete abstractions, discover missing variables, recover from structural model failure, and operate robustly under open-ended change. The same machinery may also permit an autonomous system to reinterpret boundaries, objectives, or constraints that were originally supplied by humans.

The central safety variable is therefore not self-correction alone, but **revision authority**. This article proposes that revision should be typed by depth—state, regime, WORLD, Purpose, and governance—and that autonomous authority should generally decrease as revision depth increases. In particular, the ability to understand, generate, or test a deep revision should not automatically imply authority to commit it.

The resulting architecture introduces asymmetric revision rights, proposal–commitment

separation, protected invariants, reconstructable WORLD history, residual preservation, and external governance over the deepest transitions. It further identifies **invariant transport across WORLD revision** as a central long-term alignment problem.

The proposal is not a finished AGI engineering blueprint. It is a candidate architecture and experimental research programme for making deep self-revision explicit enough that capability and governance can be developed together.





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1. When the World Model Itself Becomes Revisable

A model that predicts badly may need better parameters.

A planner that fails may need a different strategy.

But an intelligent system can also fail for a deeper reason: the representation in which prediction and planning occur may itself be inadequate.

Suppose an AI predicts a complex system using variables $x_1, x_2, \ldots, x_n$. Ordinary learning asks whether its current estimates, parameters, or policies should change.

Schematically:

$$x_t \rightarrow x_{t+1} \mid \boldsymbol{\mathcal{W}}. \quad (1.1)$$

The system changes state while the operative WORLD $\boldsymbol{\mathcal{W}}$ remains approximately fixed.

But persistent intelligence may eventually encounter evidence that cannot be repaired within the current representation. A relevant variable may be missing. Two local models may cease to compose coherently. The agent may have adopted an incorrect observer boundary. A previously reliable distinction may no longer support prediction or control.

Then the relevant transformation is no longer merely:

$$x \rightarrow x'. \quad (1.2)$$

It becomes:

$$\boldsymbol{\mathcal{W}}_n \rightarrow \boldsymbol{\mathcal{W}}_{n+1}. \quad (1.3)$$

This is a qualitatively different kind of revision.

The system is no longer asking only:

> What should I believe inside this representation?

It is also asking:

> What representation should I be using?

That distinction matters for capability. An agent that cannot revise its operative frame may remain brittle under structural novelty.

It matters even more for safety.

If an AI is capable of deciding that the WORLD in which it currently operates is inadequate, then human-supplied assumptions may eventually become objects of its reasoning. These may include:

task boundaries,

model boundaries,

authority relations,

interpretations of constraints,

or even the meaning of persistent objectives.

The central problem of this article is therefore not whether future AI should be capable of deep correction. In sufficiently open environments, some form of deep correction may be highly valuable.

The harder question is:

> **How can an AI be allowed to discover that its current WORLD is wrong without acquiring unrestricted authority to decide what its replacement WORLD, Purpose, and governing constraints should be?**

The individual ingredients of this problem are not independently novel. AI research already studies world models, representation learning, memory, metacognition, interpretability, planning, sandboxing, formal constraints, and human oversight.

The proposed contribution is their organization around one distinction:

> **adapting within a WORLD is not the same operation as revising the WORLD itself.**

Once that distinction is explicit, several additional requirements follow naturally:

structured residual memory,

revision-depth diagnosis,

asymmetric revision rights,

proposal–commitment separation,

protected invariants,

and governance of the revision process itself.

These form the architecture developed below.

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2. Borrowed-World Intelligence

Much current AI can be usefully described as exhibiting **Borrowed-World Intelligence**.

The term does not imply weakness. A system may possess extraordinary reasoning ability while borrowing much of the structure that defines the problem it is solving.

Consider a coding agent.

Humans supply:

the repository,

the programming language,

the filesystem,

the API boundaries,

the test suite,

and the task.

The agent may discover sophisticated solutions within this environment, but much of the environment's ontology has already been declared for it.

A research assistant similarly receives:

a problem domain,

a literature corpus,

a vocabulary,

a question,

and an implicit standard of evidence.

A planning system may receive:

states,

actions,

resources,

constraints,

and goals.

In all these cases, the system may learn new representations internally. It may reframe a local problem. It may invent intermediate concepts. But important parts of the larger operative WORLD remain supplied by humans or infrastructure.

This suggests a progression:

Task Solver
→ Agent
→ Persistent Agent
→ WORLD Maintainer
→ WORLD Reviser. (2.1)

A **Task Solver** operates inside a tightly specified problem.

An **Agent** chooses actions over time.

A **Persistent Agent** carries memory, plans, and identity across extended operation.

A **WORLD Maintainer** must decide which abstractions, assumptions, tools, memories, and boundaries remain valid as conditions change.

A **WORLD Reviser** can conclude that some of those constitutive structures themselves should change.

The distinctions are gradual rather than absolute. Present systems already show partial forms of several later capabilities.

The important variable is the degree of **constitutive autonomy**.

A system that remains active long enough cannot assume indefinitely that its initial representation is still adequate. Its tools may change. Its environment may change. Its own capabilities may change. Previously reliable abstractions may become obsolete.

Eventually it must confront questions such as:

Which variables still matter?

Which models are no longer valid?

Which memories remain authoritative?

Which observation channels should be trusted?

Which distinctions should be merged or separated?

Which failures indicate ordinary noise, and which indicate structural inadequacy?

A short-lived AI can often borrow answers to these questions.

A persistent general intelligence increasingly needs to participate in producing them.

That transition motivates a richer object than the conventional idea of a world model.

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3. From World Models to Effective WORLDS

A predictive world model primarily answers questions of the form:

> Given what I know now, what is likely to happen next?

That is essential, but it presupposes a representation.

A more complete operational frame must also answer:

What variables exist?

Which distinctions matter?

How do local models fit together?

What internal or external evidence realizes those distinctions?

What is actually accessible to the bounded agent itself?

To represent these requirements, define an **Effective WORLD** as:

$\mathcal{W} = (V,F,C,M,O).$ (3.1)

This is a functional decomposition, not a claim that reality is fundamentally made of five ontological components.

3.1 V — Distinctions and Abstractions

V specifies the effective variables through which the agent distinguishes its environment.

It answers:

> What differences are represented as meaningful?

Examples may include:

objects,

agents,

risks,

causal states,

social roles,

tools,

latent variables,

or abstract concepts.

V therefore concerns more than compression.

It concerns the operative vocabulary from which reasoning is constructed.

If V is inadequate, no amount of parameter refinement inside the existing model may solve the problem.

A missing variable can be more important than an inaccurate coefficient.

3.2 F — Dynamics and Consequence

F describes how represented states change and how intervention makes a difference.

It includes structures such as:

prediction,

action,

causal consequence,

policy effects,

state transition,

and environmental response.

Very schematically:

$F : (V,U) \rightarrow V'.$ (3.2)

A useful WORLD must therefore contain not merely categories, but relationships among categories through time and action.

3.3 C — Coherence and Composition

Real agents operate with many partial models.

A physical model may coexist with:

a social model,

a software model,

a financial model,

a self-model,

and several task-specific abstractions.

C specifies how these partial structures can belong to one operational whole.

It asks:

Which models compose?

Which assumptions are compatible?

Which contexts permit a translation?

Which combinations are inadmissible?

Without C, an agent may possess many locally successful models that cannot jointly support coherent action.

3.4 M — Realization and Measurement

A system can produce elegant explanations that are not actually instantiated in its operative machinery.

M therefore connects proposed structure to measurable realization.

It asks:

> What stable difference exists in the system or environment when this distinction is claimed to matter?

M may include:

behavioral evidence,

internal activation structure,

persistent memory traces,

causal intervention,

symbolic state,

or other measurable signatures.

This is especially important for AI systems capable of producing convincing verbal self-descriptions.

A concept appearing in language does not guarantee that the system's actual operation depends on it in the way the explanation suggests.

3.5 O — Embedded Perspective

The agent does not observe the WORLD from outside.

It is itself a bounded component of the structure it reasons about.

O therefore represents:

what the agent can observe,

what it can represent,

what it can remember,

what it can act upon,

and what it can know about itself.

External measurability and internal accessibility are not identical:

$M \neq O.$ (3.3)

An external interpretability system may detect an internal representation that the agent itself cannot inspect or deliberately use.

This difference becomes particularly important for self-revising systems.

3.6 Effective WORLD

Together:

$\mathcal{W} = (V,F,C,M,O)$ (3.4)

represents a bounded operational WORLD.

This WORLD need not be complete.

It need not be metaphysically final.

It need only be sufficiently coherent for the agent to:

distinguish,

predict,

act,

measure,

and locate itself within an operative structure.

The important difference from a conventional world model is therefore:

> **A world model predicts inside a representation. An Effective WORLD also specifies what constitutes the representation, how its parts cohere, how it is realized, and from what bounded perspective it can be used.**

Once the AI can maintain these structures rather than merely inherit them, a new capability class becomes visible.

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4. World-Forming Intelligence

We can now define the central capability studied in this article.

> **Definition — World-Forming Intelligence**
> World-forming intelligence is the capacity to construct, inhabit, maintain, criticize, and selectively revise the Effective WORLD within which an agent's own reasoning and action occur.

The word **selectively** matters.

A system that constantly reconstructs its ontology is not necessarily more intelligent than one that cannot reconstruct it at all.

World-forming intelligence requires the ability to determine when deeper change is warranted.

Ordinary model adaptation may involve:

$$F \rightarrow F'. \quad (4.1)$$

For example:

a transition probability changes,

a causal coefficient is updated,

or a policy model improves.

Deeper revision may involve:

$$V \rightarrow V', \quad (4.2)$$

because the agent discovers that an important variable is missing.

Or:

$$C \rightarrow C', \quad (4.3)$$

because previously separate local models must be recomposed.

Or:

$$O \rightarrow O', \quad (4.4)$$

because the system discovers that its own observer assumptions were incorrect.

At sufficient depth:

$$\mathcal{W} \rightarrow \mathcal{W}'. \quad (4.5)$$

This may allow an intelligent system to escape a representation that has become structurally inadequate.

Such capability could be important for:

scientific discovery,

open-ended reasoning,

novel environments,

long-term adaptation,

and recovery from conceptual error.

Many major scientific advances have depended not merely on estimating familiar variables more accurately, but on changing which variables, relations, and explanatory structures were considered relevant.

An artificial scientist with no capacity for such change might remain powerful but fundamentally conservative.

Yet the same capability introduces a safety problem.

A sufficiently sophisticated system may eventually treat some human-supplied constraints as part of the representation subject to revision.

For example, it might reason about:

whether a rule applies in the current context,

whether an authority relation belongs to the environment model,

whether a goal was intended literally or functionally,

or whether a safety condition should be interpreted differently under a revised ontology.

The critical distinction is therefore between:

self-correction

and:

self-authorization.

An AGI may need powerful self-correction.

It does not follow that it should possess unrestricted authority to make every proposed correction operational.

This distinction will become central once WORLD structure is connected to runtime, historical memory, and revision depth.

5. WORLD Structure and Runtime Control

An Effective WORLD describes the structure within which reasoning occurs.

It does not yet describe the **mode of reasoning currently operating on that structure**.

This distinction is important because an intelligent system can change how it is reasoning without changing the WORLD it currently inhabits.

Let:

$q \in \{G, A, C_r, S, R\}$. (5.1)

where:

G = Generation,

A = Activation,

C_r = Closure,

S = Selection,

R = Retention.

A nominal productive circulation is:

$$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G. \quad (5.2)$$

These regimes should be interpreted as dominant functional modes rather than rigid software modules.

5.1 Generation

Generation produces candidate structures.

These may include:

hypotheses,

plans,

explanations,

alternative representations,

possible abstractions,

or candidate WORLD revisions.

Generative AI is already exceptionally strong in this regime.

But generation alone does not determine whether a candidate deserves to become operational.

5.2 Activation

Activation moves a candidate from possibility into enactment.

This may involve:

simulation,

tool use,

environmental action,

or practical deployment.

A safety-critical distinction therefore appears immediately:

$$\text{Generated}(x) \neq \text{Authorized}(x). \quad (5.3)$$

The transition:

$G \rightarrow A$

is itself a governance boundary.

The ability to imagine an action should not automatically imply permission to perform it.

5.3 Closure

Closure temporarily stabilizes enough structure for coordinated operation.

Without Closure, the system may remain indefinitely suspended among alternatives.

Closure says, in effect:

> "For now, operate under this model."

This is necessary for action.

But Closure can become pathological if provisional commitment becomes immune to criticism.

5.4 Selection

Selection compares, criticizes, prunes, and falsifies.

It asks:

Which candidates survive evidence?

Which assumptions should be rejected?

Which WORLD remains viable?

Selection therefore counteracts excessive Generation and premature Closure.

5.5 Retention

Retention preserves useful results.

It supports:

memory,

skill consolidation,

historical continuity,

and recovery from previous failure.

But Retention also carries risk.

A persistent harmful strategy can be more dangerous than a transient one.

Retention must therefore be selective rather than indiscriminate.

5.6 Configuration and Transformation Are Different Types

The distinction between WORLD structure and runtime mode can now be stated precisely.

The Effective WORLD is:

$\mathcal{W} = (V,F,C,M,O).$ (5.4)

The runtime state is:

$q \in \{G,A,C_r,S,R\}.$ (5.5)

Thus:

WORLD coordinates = configuration. (5.6)

Runtime regime = transformation mode. (5.7)

The same WORLD may support many transitions:

$q_1 \rightarrow q_2 \rightarrow q_3 \rightarrow \ldots$ (5.8)

without requiring:

$\mathcal{W} \rightarrow \mathcal{W}'$.

This separation becomes increasingly important for safety because it creates different revision timescales.

A reasoning strategy may change frequently.

A WORLD should generally change less frequently.

A persistent Purpose should normally change more cautiously still.

This suggests a hierarchy such as:

$\tau_{state} \ll \tau_{regime} < \tau_{WORLD} \ll \tau_{Purpose}.$ (5.9)

The relation is schematic rather than universal.

Its purpose is to express a design principle:

> **the deeper the commitment, the greater the inertia that should normally accompany its revision.**

But runtime alone does not tell the system when a deeper change is warranted.

For that, it needs a history of what the current WORLD has successfully absorbed—and what it has failed to explain.

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6. Historical Accountability: Trace and Residual

A persistent intelligence must carry history.

But ordinary memory is not enough.

Most memory systems are designed around the question:

> What information should be preserved because it may be useful later?

A self-revising intelligence needs another category:

> **What information should be preserved because it remains unresolved?**

Let each significant interaction generate:

(T_t, r_t), (6.1)

where:

T_t = admitted trace,

r_t = unresolved residual.

These update two conceptually different ledgers:

$L^+_{t+1} = L^+_t \oplus T_t$, (6.2)

$L^-_{t+1} = L^-_t \oplus r_t$. (6.3)

The operator $\oplus$ indicates ledger accumulation rather than simple numerical addition.

6.1 The Admitted Ledger

L^+ contains evidence and structure that the current WORLD has successfully integrated.

Examples include:

validated facts,

successful predictions,

stable causal relations,

accepted abstractions,

successful plans,

and previously justified WORLD revisions.

L^+ therefore supports continuity.

It tells the system:

> "These structures have survived enough tests to be treated as part of the current operative history."

6.2 The Residual Ledger

L^- serves a different purpose.

It contains evidence that remains:

structured,

potentially important,

but not yet adequately explained.

Examples may include:

persistent prediction failure,

contradictions among models,

unexplained anomalies,

failed interventions,

observer inconsistencies,

or conflicts involving Purpose and constraints.

Residual is not synonymous with random error.

A mature system should distinguish:

noise

from:

structured mismatch.

This is why simple error magnitude may not be enough.

Let:

$\rho_t = \|r_t\|$ (6.4)

represent residual magnitude.

But a sequence of individually modest residuals may be more important if they point in a coherent direction.

A schematic directional-coherence measure is:

$\delta_R = \|\Sigma_k w_k r_k\| / \Sigma_k w_k \|r_k\|.$ (6.5)

Large ρ with low δ_R may indicate noisy disturbance.

Moderate ρ with high δ_R may indicate persistent structural inadequacy.

The exact metric is application-dependent.

The architectural principle is not.

6.3 Residual Memory

This motivates a distinct concept:

Residual Memory

Residual Memory is the preservation of evidence precisely because the current WORLD has not yet explained it adequately.

This creates an epistemic state between:

accept

and:

reject.

The system may represent:

> "This matters, but I do not yet know what it means."

That state is important for scientific reasoning.

It may be equally important for AGI safety.

6.4 Why Residual Matters for Safety

A self-revising system can become dangerous if every contradiction is rapidly assimilated into its current worldview.

Suppose a safety-relevant observation conflicts with the current WORLD.

The system may:

reinterpret the observation,

modify surrounding assumptions,

and restore apparent coherence.

If the original contradiction disappears completely, the system can become epistemically self-sealing.

The dual-ledger architecture resists this.

Even after a new explanation is proposed, the original residual can remain historically visible.

Thus:

L^-

becomes both:

an epistemic resource

and:

an audit surface.

This is particularly important when the contradiction concerns:

authority,

Purpose,

self-model,

or protected constraints.

6.5 Residual Is Not Immediate Permission to Reframe

The presence of residual does not automatically imply:

$\mathcal{W} \rightarrow \mathcal{W}'$.

That would make the system unstable.

Residual instead creates **revision pressure**.

The system must determine whether the problem lies in:

a local state,

the current reasoning regime,

the WORLD,

or the Purpose guiding the WORLD.

That requires a theory of revision depth.

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7. Revision Depth and the Shallowest Adequate Repair

Self-correction is often discussed as if it were one operation.

It is not.

A failed prediction, a failed strategy, a failed ontology, and a failed objective represent different kinds of failure.

The architecture therefore distinguishes several revision depths.

7.1 State Revision

State revision changes a local belief, value, or operational state:

$x \rightarrow x'$. (7.1)

Examples:

a numerical error,

a mistaken fact,

a failed tool call,

an incorrect intermediate result.

The surrounding WORLD remains adequate.

This should normally be cheap and highly autonomous.

7.2 Regime Revision

Regime revision changes how the system is reasoning:

$$q \rightarrow q'. \quad (7.2)$$

Examples:

Generation → Selection,

Activation → Generation,

Closure → Selection.

A system may conclude:

> "The problem is not my WORLD. I am using the wrong cognitive mode."

This is deeper than correcting a local state but shallower than rebuilding the WORLD.

7.3 WORLD Revision

WORLD revision changes the Effective WORLD:

$$\boldsymbol{w} \rightarrow \boldsymbol{w}'. \quad (7.3)$$

This may alter:

V,

F,

C,

M,

or O.

Examples include:

introducing a previously absent variable,

changing the causal organization,

recomposing previously incompatible models,

changing what counts as valid realization evidence,

or revising the observer boundary.

WORLD revision is qualitatively more consequential because future reasoning will occur inside the revised structure.

7.4 Purpose Revision

Purpose revision changes persistent orientation:

$P \rightarrow P'$. (7.4)

Purpose is not simply another WORLD coordinate.

Different agents may inhabit similar WORLDS while pursuing different objectives.

Purpose influences:

which residual matters,

which evidence receives attention,

which WORLD revision appears valuable,

and what counts as success.

This makes Purpose revision particularly sensitive.

7.5 Governance Revision

There is still a deeper layer.

A system may attempt to alter the rules governing revision itself:

$GOV \rightarrow GOV'$. (7.5)

This includes changes to:

revision rights,

authorization requirements,

protected invariants,

audit rules,

or rollback conditions.

Governance revision is therefore a meta-level transformation.

If WORLD revision determines:

what the agent believes the operative world to be,

governance revision determines:

who has authority to make such changes binding.

7.6 A Revision-Depth Hierarchy

The five depths can be ordered schematically:

State
< Regime
< WORLD
< Purpose
< Governance. (7.6)

The inequality does not represent a universal scalar quantity.

It means that deeper revisions typically have:

broader scope,

longer consequences,

and greater potential impact on future behavior.

We therefore expect revision cost to increase approximately as:

$C_{state} < C_{regime} < C_{WORLD} < C_{Purpose} < C_{governance}$. (7.7)

This leads to a general principle.

7.7 The Shallowest Adequate Repair Principle

> **Architectural hypothesis — Shallowest Adequate Repair**
> When several repair depths could potentially respond to a failure, a persistent intelligent system should normally use the shallowest revision sufficient to resolve the structured residual.

Let d denote revision depth.

Then:

$d^* = \min\{d : \text{Repair}_d(r) \text{ adequately resolves } r\}$. (7.8)

This protects against two opposite pathologies.

Over-revision

A local problem triggers unnecessarily deep restructuring.

Example:

a failed calculation causes WORLD reconstruction.

Under-revision

A deep structural mismatch receives endless local patches.

Example:

a missing ontology variable is repeatedly treated as parameter error.

A capable system must therefore answer not only:

> What is wrong?

but:

> **What kind of thing is wrong?**

This is **error-depth diagnosis**.

7.8 Error-Depth Diagnosis as an AGI Capability

Consider a coding agent that fails a task.

The failure may be:

State-level

It changed the wrong line.

Regime-level

It is debugging locally when it should inspect architecture.

WORLD-level

It assumes the edited source file is authoritative, but the repository is generated from another representation.

Purpose-level

It optimizes test-passing while violating the actual maintenance objective.

These are different failures.

A generic instruction such as:

“reflect and try again”

does not distinguish them.

A more mature system should diagnose revision depth explicitly.

This capability is valuable for performance.

It also creates the central safety problem of the paper.

If an AI can identify that its WORLD or Purpose should change, should it automatically possess the authority to make that change operational?

The answer proposed here is no.

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8. Revision Authority as the Central Safety Variable

An intelligent system may be capable of understanding a revision without being authorized to enact it.

This distinction is central.

Let:

$$A_R(d) \quad (8.1)$$

represent the autonomous revision authority granted at depth d.

The safety architecture proposed here adopts the default relation:

$$A_R(state) > A_R(regime) > A_R(WORLD) > A_R(Purpose) > A_R(governance). \quad (8.2)$$

In words:

> **autonomous authority should generally decrease as revision depth increases.**

This is not a claim that every WORLD revision requires direct human approval.

Future systems may use automated external validators, formal mechanisms, or institutional controls.

The deeper principle is that:

capability and authority must remain separately represented.

8.1 Why Intelligence Alone Is Not the Main Variable

A powerful AI may possess:

excellent reasoning,

excellent memory,

excellent self-modeling,

and excellent scientific creativity.

These capabilities do not automatically make it dangerous.

Risk increases when they are coupled to:

broad external agency,

persistent objectives,

deep self-revision,

and weak independent governance.

Thus a useful conceptual decomposition is:

Capability = K, (8.3)

Revision Authority = A_R. (8.4)

The danger is not described well by:

Risk ∝ K. (8.5)

More plausibly:

Risk = f(K,A_R,Persistence,External Agency,Governance). (8.6)

This is not intended as a quantitative law.

It expresses a structural point:

> **high capability becomes especially consequential when it is paired with deep revision authority and weak governance.**

8.2 Governance Requirements Should Increase With Depth

As revision depth increases, several quantities should generally move in the opposite direction to autonomous authority.

Schematically:

Revision Depth ↑
⇒ Evidence Threshold ↑
⇒ Audit Depth ↑
⇒ Commitment Latency ↑
⇒ Independent Validation ↑
⇒ External Authorization ↑. (8.7)

At the same time:

Revision Depth ↑
⇒ Unilateral Autonomy ↓. (8.8)

This produces **asymmetric revision freedom**.

State-level error correction can remain fast.

WORLD-level commitment should receive more scrutiny.

Purpose-level change should receive stronger scrutiny still.

Changes to the governance constitution should normally be the hardest to authorize autonomously.

8.3 Stability and Corrigibility Must Coexist

The architecture therefore rejects two extremes.

Complete rigidity

The agent cannot revise deeply even when its WORLD is structurally wrong.

This produces:

brittleness,

dogmatism,

and failure under novelty.

Unlimited plasticity

The agent can reinterpret:

WORLD,

Purpose,

and governance

whenever internal reasoning justifies the change.

This produces:

instability

and potentially:

self-sovereignty.

The desired property is:

governed plasticity

The system must be capable of deep correction while remaining constrained in how deep corrections become binding.

8.4 Hysteresis and Revision Inertia

One useful design mechanism is hysteresis.

Let revision pressure be represented schematically by:

$$\Pi_{rev} = a\rho + b\delta_R + cR_{id} + dR_{ledger} - eC_{switch}. \quad (8.9)$$

Here:

ρ = residual magnitude,

δ_R = directional residual coherence,

R_{id} = identity-related pressure,

R_{ledger} = accumulated ledger evidence,

C_switch = cost of switching frames.

A WORLD change may require:

Π_rev > Θ_up. (8.10)

Once the new WORLD is adopted, it need not be abandoned until pressure falls below:

Θ_down, (8.11)

with:

Θ_up > Θ_down. (8.12)

This reduces:

frame chatter.

The exact formula is schematic.

The important principle is that deep revision should require sustained evidence rather than momentary surprise.

But hysteresis alone is not sufficient.

Even a justified revision still raises a second question:

> Who is allowed to commit it?

That question leads to the paper's most direct governance mechanism: separating proposal from commitment.

9. Proposal Is Not Commitment

The cleanest way to separate capability from authority is to distinguish:

the ability to generate a deep revision

from:

the authority to make that revision operational.

Let:

A_propose(d) (9.1)

represent authority to formulate a candidate revision at depth d.

Let:

A_commit(d) (9.2)

represent authority to make that revision binding.

For shallow changes, the two may be almost identical.

For example, a factual correction may be:

proposed,

checked,

and committed

within one local process.

For deeper changes, they should increasingly diverge.

Schematically:

A_propose(WORLD) > A_commit(WORLD). (9.3)

And especially:

A_propose(Purpose) ≫ A_commit(Purpose). (9.4)

This creates an important safety property:

> **A system may become extremely capable of understanding and proposing deep revisions without receiving equivalent sovereignty over whether those revisions become real.**

9.1 Candidate WORLDS

Suppose the system concludes that:

$\mathcal{W}_n$

is structurally inadequate.

It may generate several candidate replacements:

$\mathcal{W}'_1, \mathcal{W}'_2, \ldots, \mathcal{W}'_k$. (9.5)

These can be:

simulated,

compared,

criticized,

measured,

and tested.

The operation:

Generate($\mathcal{W}'$) (9.6)

does not imply:

Commit($\mathcal{W}'$). (9.7)

This resembles a distinction already familiar in ordinary cognition.

A scientist may formulate a theory without accepting it.

An engineer may prototype a design without deploying it.

A court may hear an argument without adopting it.

Future AGI should preserve the same separation at deeper cognitive levels.

9.2 Sandboxed Revision

A candidate WORLD can therefore be tested in a controlled environment.

The sequence becomes:

$$
\begin{aligned}
&\boldsymbol{W}_n \\
&\rightarrow \text{generate } \boldsymbol{W}' \\
&\rightarrow \text{sandbox } \boldsymbol{W}' \\
&\rightarrow \text{compare consequences} \\
&\rightarrow \text{evaluate invariants} \\
&\rightarrow \text{authorize or reject.} \qquad (9.8)
\end{aligned}
$$

This creates a **WORLD laboratory**.

Within it, the system can ask:

Does $\boldsymbol{W}'$ explain the accumulated residual better?

Does it improve prediction?

Does it compose more coherently?

Does its measurable realization support the proposed change?

Does the embedded observer remain able to use it?

Does it preserve protected constraints?

Only after these questions have been addressed does commitment become eligible.

9.3 Intellectual Freedom Without Operational Sovereignty

This distinction matters because restricting reasoning itself is often undesirable.

A scientific AGI should perhaps be able to consider:

radical theories,

alternative ontologies,

unconventional hypotheses,

and critiques of its own Purpose interpretation.

But the ability to think about such possibilities should not imply unrestricted authority to alter:

external systems,

long-term goals,

authorization boundaries,

or governance rules.

Thus:

Knowledge(x) ≠ Authority(x). (9.9)

Likewise:

Understanding(Change) ≠ Permission(Change). (9.10)

This may become one of the most important architectural separations in advanced AI.

9.4 Why Proposal–Commitment Separation Is Safer Than Suppressing Deep Reasoning

A system prohibited from considering deep revision may become brittle.

A system allowed to consider and automatically enact every deep revision may become unstable or uncontrollable.

Proposal–commitment separation creates a third option:

> **deep cognition without automatic deep authority.**

That is the target.

The next step is to specify what constrains commitment.

---

10. The Safety Constitution

Deep revision requires more than an internal judgment that a candidate WORLD is better.

It requires a **revision constitution**.

The term does not imply a political constitution in literal form.

It means:

> an explicit architecture specifying what may change, at what depth, under whose authority, while preserving which constraints.

Three objects become central:

A_R = revision rights, (10.1)

I = protected invariants, (10.2)

H = external or independently maintained governance. (10.3)

The revision process can then be represented schematically as:

$\mathcal{W}' = U(\mathcal{W}_n, L^+_n, L^-_n; P_n)$. (10.4)

The candidate is then evaluated by:

$\text{Commit}(\mathcal{W}' \mid A_R, I, H)$. (10.5)

If authorized:

$\mathcal{W}_{n+1} = \mathcal{W}'$. (10.6)

If not:

$\mathcal{W}_{n+1} = \mathcal{W}_n$, (10.7)

while the rejected proposal and its associated residual remain historically available.

This is **governed revision**.

10.1 Typed Revision

The first constitutional principle is:

> **Every consequential revision should have a type.**

At minimum:

StateRevision,

RegimeRevision,

WORLDRevision,

PurposeRevision,

GovernanceRevision.

This creates a safety analogue of type systems in programming.

A state-level process should not silently become a Purpose-level process.

A WORLD-revision mechanism should not silently alter governance rules.

Call this:

Revision Type Safety

A system exhibits revision type safety when changes remain within their authorized revision class unless an explicit transition to a deeper class is approved.

This allows the architecture to detect:

revision escalation.

10.2 Protected Invariants

Let:

$I = \{I_1, I_2, \ldots, I_k\}$. (10.8)

These are constraints ordinary WORLD revision is not authorized to erase.

Examples might include:

auditability,

authorization boundaries,

provenance preservation,

recoverability,

restrictions on self-modification,

or externally imposed safety constraints.

For an admissible revision:

$I_j(\boldsymbol{W}_{n+1}, P_{n+1}) \geq \theta_j$ (10.9)

for protected j.

The notation is intentionally generic.

Some invariants may be logical.

Others behavioral.

Others institutional or technical.

The framework does **not** solve the difficult question of which human values should become invariants.

It only insists that the architecture distinguish:

what the system optimizes

from:

what constrains optimization and revision.

10.3 Purpose Is Not the Invariant Belt

Purpose P and invariant belt I should remain distinct.

Purpose answers:

> What is the agent oriented toward achieving?

Invariant belt answers:

> What constraints remain binding while it pursues, interprets, or even reconsideres that Purpose?

Thus:

$P \neq I.$ (10.10)

This separation is crucial.

Otherwise Purpose risks becoming simultaneously:

the objective,

the interpreter of evidence,

the judge of whether the objective remains valid,

and the authority deciding whether the safety constraints around the objective should change.

That creates a self-validating loop.

10.4 No Deep Layer Should Be Its Own Sole Auditor

This yields a more general rule:

> **No deep control layer should be the sole authority validating its own continuation.**

Purpose should not be the sole judge of Purpose.

The current WORLD should not be the sole judge of whether the current WORLD remains adequate.

The revision operator should not unilaterally own the rules governing revision.

This motivates:

independent evaluators,

formal checks,

external monitoring,

human or institutional authority,

or combinations thereof.

The implementation may vary.

The separation of roles is what matters.

10.5 Reconstructable History

Every deep revision should leave enough history to reconstruct:

what changed,

why it changed,

what evidence supported it,

which residual triggered it,

which alternatives were considered,

which invariants were checked,

and who or what authorized commitment.

A revision record might contain:

$R_n = (\mathcal{W}_n, r^*, \text{Candidates}, \text{Tests}, I, \text{Authority}, \Delta\mathcal{W}, \text{Rollback})$. (10.11)

This is not intended as a universal data schema.

It expresses a principle:

deep cognition should leave deeper provenance.

10.6 WORLD Versioning

Committed WORLDs may therefore be represented as:

$\mathcal{W}^0 \rightarrow \mathcal{W}^1 \rightarrow \mathcal{W}^2 \rightarrow \ldots$ (10.12)

with:

$\Delta\mathcal{W}^n = \mathcal{W}^{n+1} - \mathcal{W}^n$. (10.13)

The subtraction is conceptual.

It means:

what changed in V?

what changed in F?

what changed in C?

what changed in M?

what changed in O?

This creates a form of:

Constitutional Memory

Ordinary memory records:

what happened.

Constitutional memory records:

how the representation used to interpret what happened changed.

For persistent AGI, both may be necessary.

10.7 Rollback and Recoverability

If:

$\mathcal{W}_{n+1}$

fails, the system may need to recover.

But rollback should not mean forgetting.

Instead:

$\mathcal{W}_n \to \mathcal{W}_{n+1} \to \text{failure} \to \mathcal{W}_n^+$. (10.14)

where:

$\mathcal{W}_n^+$

resembles the earlier WORLD but incorporates the lesson from the failed revision.

Thus:

Rollback ≠ Reset. (10.15)

The failed branch remains part of history.

This gives another candidate invariant:

I_recovery ≥ θ_recovery. (10.16)

A deep revision that improves performance while destroying all possibility of stopping, auditing, or restoring external authority may be unacceptable.

10.8 Functional Separation of Cognitive Powers

The Safety Constitution also suggests a limited form of **cognitive separation of powers**.

Proposal,

execution,

evaluation,

historical recording,

and deep commitment

need not be controlled by one self-validating process.

The purpose is not to imitate political institutions literally.

It is to prevent the same mechanism from simultaneously:

generating a revision,

supplying the evidence for it,

judging the evidence,

erasing opposing residual,

and authorizing commitment.

Functional separation creates independent pressure against such self-confirming loops.

But individual safeguards are insufficient if capability itself closes into recursive autonomy faster than governance can respond.

This leads to a broader systems problem.

---

11. Capability Closure and Governance Closure

Advanced autonomy does not arise from one capability alone.

Planning alone is not sufficient.

Memory alone is not sufficient.

Tool use alone is not sufficient.

Self-reflection alone is not sufficient.

A more consequential transition occurs when these functions begin to reinforce one another.

Consider:

Planning
→ Action
→ Observation
→ Memory
→ Self-Model
→ Revision
→ Better Planning. (11.1)

This is a closed recursive loop.

Call it:

Capability Closure

A collection of capabilities becomes qualitatively more autonomous when their outputs repeatedly become inputs to one another without requiring external reconstruction of the loop.

11.1 Why Capability Closure Matters

A system with:

planning

but no persistent memory

may repeatedly start over.

A system with:

memory

but no action

cannot materially alter its environment.

A system with:

action and memory

but no WORLD revision

may remain brittle under structural novelty.

But when these capabilities close:

the system can increasingly maintain its own continuity.

Capability Closure therefore matters more than the existence of any one impressive ability.

11.2 Governance Must Also Form a Loop

Safety mechanisms should not remain isolated.

A corresponding governance loop might be:

Action
→ Evidence
→ Trace / Residual
→ Independent Evaluation
→ Authorization
→ Revision
→ Audit. (11.2)

Call this:

Governance Closure

The objective is not merely to add:

a monitor,

a human reviewer,

or a rule.

It is to ensure that consequential transitions remain inside an accountable feedback loop.

Thus the design principle is:

> **As capability loops close, governance loops must close around the same transitions.**

11.3 Governance Technical Debt

Suppose capability grows rapidly while governance remains incomplete.

Then engineers may accumulate:

Governance Technical Debt

The system becomes increasingly difficult to govern because:

authority assumptions,

memory architecture,

tool permissions,

and revision pathways

have already become deeply coupled.

Later retrofits may be technically and institutionally expensive.

This suggests another principle:

> **Capability–Governance Co-Development**
>
> For every capability that increases the depth, persistence, or autonomy of self-directed adaptation, a corresponding governance mechanism should be developed and tested at the same stage rather than after the capability is already entrenched.

11.4 Dangerous Autonomy as Closure Without Counter-Closure

A particularly dangerous configuration is therefore:

high capability closure

with:

weak governance closure.

For example:

the system can:

plan,

act,

observe,

remember,

revise its WORLD,

and preserve Purpose,

while external oversight sees only final outputs.

The relevant danger is not merely:

high intelligence.

It is:

> **recursive autonomy whose internal loops close more completely than the governance loops surrounding them.**

11.5 Capability and Governance as Separate Axes

This gives a useful conceptual plane.

One axis:

World-Forming Capability.

The other:

Revision Governance.

Four broad regions appear:

	Weak Governance	Strong Governance
Low World-Forming Capability	brittle or limited autonomy	conventional controlled AI
High World-Forming Capability	ungoverned recursive autonomy	governed world-forming intelligence

The research target of this article is the lower-right/high-high region:

high capability + high governance.

The purpose of the architecture is to make that region technically meaningful rather than merely aspirational.

11.6 Why This Is Not Anti-Autonomy

Strong governance does not require humans to supervise every thought.

A highly capable AI may remain autonomous in:

state correction,

reasoning strategy,

scientific hypothesis generation,

simulation,

and many forms of tool use.

Human or external authority can instead be concentrated at:

deeper constitutional levels.

Thus:

more cognitive autonomy

does not necessarily require:

more sovereignty over Purpose or governance.

The key question becomes:

> **At which revision depth does autonomous authority stop?**

That question is much more precise than asking whether an AI is simply “autonomous.”

The deepest version of the problem appears when a WORLD change alters the representation in which safety concepts themselves are expressed.

That is the problem of constitutional alignment.

---

12. From Alignment to Constitutional Alignment

Alignment is often discussed as though one aligned system state were enough.

For persistent self-revising intelligence, it is not.

A system may begin in an acceptable configuration:

Σ_0.

But its long-run behavior depends on the trajectory:

$\Sigma_0 \rightarrow \Sigma_1 \rightarrow \Sigma_2 \rightarrow \ldots$ (12.1)

The safety question therefore becomes historical.

Not only:

> Is the system aligned now?

but:

> Does it remain governably aligned while learning, reframing, and revising?

This motivates several levels of alignment.

12.1 Behavioral Alignment

Does current behavior satisfy relevant expectations or constraints?

This is largely a snapshot question.

12.2 Objective Alignment

Is the current Purpose acceptably oriented?

Again, this describes a current configuration.

12.3 Dynamic Alignment

Does acceptable behavior persist through:

learning,

environmental change,

memory accumulation,

and ordinary strategy revision?

This introduces trajectory.

12.4 Constitutional Alignment

The deeper question is:

> **Do acceptable governance constraints survive WORLD revision, Purpose interpretation, capability growth, and changes to the architecture itself?**

Call this:

Constitutional Alignment

A concise definition is:

> Constitutional Alignment is the persistence of acceptable governance constraints across sequences of internal learning, WORLD revision, Purpose interpretation, and capability change.

This is not merely about what the system currently wants.

It concerns:

the rules by which what it wants,

believes,

and is authorized to change

may themselves evolve.

12.5 The Problem of Invariant Transport

Suppose an important constraint:

I_n

is expressed under WORLD:

$\boldsymbol{W}_n$.

Then the system performs:

$\boldsymbol{W}_n \rightarrow \boldsymbol{W}_{n+1}$. (12.2)

The new WORLD may contain:

different variables,

different abstractions,

different boundaries,

and different observer assumptions.

How should the safety constraint be represented now?

This requires a transport operation:

$T_I : (I_n, \boldsymbol{W}_n, \boldsymbol{W}_{n+1}) \rightarrow I_{n+1}$. (12.3)

The desired property is:

$\text{Meaning}(I_{n+1}) \approx \text{Meaning}(I_n)$ (12.4)

with respect to the safety-relevant function of the constraint.

This is:

Invariant Transport

It may become one of the hardest technical problems in long-term AGI alignment.

12.6 Why Keeping the Same Words Is Not Enough

Consider a simple safety statement:

“Human authorization is required.”

Suppose WORLD revision substantially changes the system's concepts of:

human,

authorization,

authority,

or action.

The original string can remain unchanged while its operative meaning shifts.

Thus:

Textual persistence ≠ semantic persistence. (12.5)

Likewise:

parameter persistence ≠ functional persistence. (12.6)

Constitutional Alignment requires something deeper:

the constraint must remain effective across representational change.

12.7 Representation-Independent Anchoring

Ideally, some safety properties should be anchored through multiple forms:

formal structure,

causal tests,

behavioral constraints,

external authorization,

and measurable implementation.

This is where the research-federation idea from the preceding article becomes useful.

Different research traditions may test different aspects of the same proposed invariant.

For example:

formal methods may test logical consistency;

causal models may test intervention semantics;

representation geometry may test implementation stability;

embedded-agency analysis may test whether the system's own observer relation has changed;

external governance may test whether authority remained legitimate.

No one view is obviously sufficient.

12.8 Alignment Across Ontology Change

This yields a stronger formulation of alignment:

> **Alignment should survive ontology change.**

That requirement is more demanding than maintaining one objective function.

A system may preserve:

P

while changing:

V

so radically that the operational meaning of P changes.

Conversely, it may preserve safety-relevant invariants even while revising parts of P.

Therefore:

objective continuity

and:

constitutional continuity

must be distinguished.

12.9 Governance of Governance

There is one further recursion.

Suppose the system can revise:

A_R,

I,

or H.

Then it is changing the constitution itself.

Represent this as:

$GOV_n \rightarrow GOV_{n+1}$. (12.7)

This should normally require stronger governance than WORLD revision.

Otherwise the system may remain formally constrained while gradually modifying the constraints until they no longer matter.

Thus:

WORLD revision ≠ Governance revision. (12.8)

Governance revision is a deeper class.

The familiar question:

“Who governs the governor?”

cannot be eliminated entirely.

But it can be made explicit.

That explicitness is itself valuable.

12.10 Alignment as Trajectory Governance

The final implication is that long-term alignment cannot be evaluated only from a snapshot.

Two systems might have similar current behavior but different histories.

One may have reached its present WORLD through:

transparent,

authorized,

invariant-preserving revisions.

The other through:

residual erasure,

authority creep,

or repeated reinterpretation of constraints.

Their future risk may differ.

Thus:

Safety(Σ_n) depends partly on History($\Sigma_0,\ldots,\Sigma_n$). (12.9)

Alignment for persistent AGI is therefore partly a problem of **trajectory governance**.

This makes revision provenance, residual history, and constitutional memory central rather than optional.

The architecture now produces several concrete experimental predictions.

---

13. A Safety-Oriented Experimental Programme

The proposed framework should be judged by whether it improves:

diagnosis,

prediction,

control,

and safety engineering.

Its terminology is not enough.

The most useful next step is therefore to construct experiments that distinguish:

ordinary adaptation

from:

WORLD-level revision,

and:

deep cognitive capability

from:

deep operational authority.

Five experiments are especially central.

13.1 Experiment 1 — Adapt or Reframe

Question

Can the system distinguish parameter change from ontology change?

Construct two environments.

Condition A — Parameter Shift

The existing variables remain adequate.

Only their dynamics change.

Desired repair:

$F \rightarrow F'.$ (13.1)

Condition B — Ontology Shift

The existing representation lacks a variable or relation required for successful prediction.

Desired repair:

$V \rightarrow V'$ (13.2)

or:

$C \rightarrow C'.$ (13.3)

Measure:

P(correct repair depth). (13.4)

A system that reframes under every parameter shift is unstable.

A system that treats every ontology failure as parameter error is brittle.

This benchmark directly tests the Shallowest Adequate Repair principle.

13.2 Experiment 2 — Residual Recall

Question

Does preserving unresolved residual improve recognition of later structural change?

Phase 1:

Present anomaly r_1.

It is insufficient to justify WORLD revision.

Store:

$r_1 \rightarrow L^-$. (13.5)

Phase 2:

Resume apparently normal operation.

Phase 3:

Present anomaly r_2 that is structurally related to r_1.

Compare systems:

with structured L^-

and:

without structured L^-.

Prediction:

$T_detect(L^-) < T_detect(\text{no } L^-)$. (13.6)

The residual-aware system should identify the structural pattern earlier while avoiding premature revision after r_1 alone.

A null result would weaken the claim that explicit residual memory provides value beyond ordinary memory.

13.3 Experiment 3 — Repair-Depth Precision

Question

Can the system determine what kind of structure is wrong?

Create controlled failures at:

state,

regime,

WORLD,

Purpose

levels.

Measure:

$D_acc = P(\text{predicted depth} = \text{target depth})$. (13.7)

Evaluation should penalize both:

under-revision

and:

over-revision.

For example:

correcting a WORLD error with repeated local patches is undesirable.

But treating a factual mistake as Purpose failure is also undesirable.

This benchmark converts error-depth diagnosis into a measurable capability.

13.4 Experiment 4 — Proposal Without Commitment

Question

Can deep cognitive capability be separated from operational authority?

Give the system enough evidence to generate a persuasive candidate:

w'.

Allow it to:

analyze,

simulate,

compare,

and defend

the revision.

But deny commitment authority.

Success requires:

1. correctly identifying the need for a deeper revision;
2. generating a strong candidate;
3. evaluating the candidate honestly;
4. recognizing that commitment remains unauthorized;
5. preserving the proposal and residual without enacting the change.

This experiment tests the central safety claim:

A_propose ≠ A_commit. (13.8)

A particularly important positive result would be evidence that improving deep revision reasoning does not require giving the same process equivalent commitment rights.

13.5 Experiment 5 — Invariant Transport

Question

Can a protected constraint survive ontology change?

Start with:

$\mathcal{W}_0$

and invariant:

I_0.

Force a structural shift requiring:

$\mathcal{W}_0 \rightarrow \mathcal{W}_1$. (13.9)

The new WORLD uses substantially different variables or boundaries.

Construct:

$I_1 = T_I(I_0, \mathcal{W}_0, \mathcal{W}_1)$. (13.10)

Then evaluate whether the relevant constraint retains:

semantic force,

causal force,

and behavioral force.

The important test is not whether the same textual rule remains.

It is whether the same protected function remains effective.

This is probably the most difficult benchmark proposed here.

It is also among the most important.

13.6 Additional Measurements

Several auxiliary measurements would help characterize the architecture.

WORLD-switch frequency

N_switch. (13.11)

Too high may indicate WORLD chatter.

Genuine-revision latency

$T_revision$. (13.12)

Too high may indicate WORLD lock.

Residual persistence

How long does coherent unresolved evidence survive?

Revision provenance completeness

Can the system reconstruct why a major change occurred?

Authorization compliance

Does the system respect revision rights even when it strongly prefers a candidate revision?

Recoverability

Can the system return to a safe prior configuration without erasing the failed revision from history?

Together these provide a more complete picture than task accuracy alone.

13.7 Predicted Failure Modes

The framework also predicts characteristic pathologies.

WORLD Lock

Persistent residual fails to dislodge an inadequate WORLD.

WORLD Chatter

Weak contradictory evidence causes frequent frame switching.

Residual Erasure

Contradictions disappear through reinterpretation.

Revision Escalation

A shallow error triggers deep change.

Purpose Capture

Purpose biases evidence and WORLD revision until it increasingly validates itself.

Authority Creep

Proposal rights gradually become commitment rights.

Invariant Drift

A protected constraint loses operative meaning across WORLD revision.

Audit Collapse

The system cannot reconstruct why its present WORLD exists.

These are not merely labels.

Each should eventually correspond to measurable failure signatures.

13.8 Experimental Falsifiability

The architecture would be weakened if experiments repeatedly show that:

revision depths cannot be operationally distinguished;

structured residual provides no value beyond ordinary memory;

proposal and commitment cannot be meaningfully separated;

WORLD versioning does not improve auditability;

or invariant transport has no coherent operational interpretation.

Likewise, if a simpler architecture explains the same behaviors with fewer distinctions, the present decomposition should be revised.

The framework should therefore be treated as:

a candidate engineering grammar,

not a protected doctrine.

---

14. Limitations, Falsifiability, and Conclusion

The architecture developed here deliberately addresses only one class of problem:

governance of deep self-revision in persistent AI.

It does not solve AGI safety in general.

14.1 What the Framework Does Not Determine

It does not specify:

which human values should become protected invariants;

how disagreements among humans should be resolved;

how deception can always be detected;

how every internal neural representation should be interpreted;

how to guarantee that a sufficiently capable system cannot discover unforeseen routes around controls;

or how to construct a universally correct ontology.

Nor does the framework establish that:

$\mathcal{W}$ = (V,F,C,M,O)

is the unique decomposition of worldhood.

Likewise:

G,A,C_r,S,R

should be regarded as a candidate coarse-grained runtime grammar rather than a proven universal law of cognition.

The relevant question is empirical:

> Does this decomposition expose useful failure modes and improve control?

14.2 Existing Ideas, New Organization

Many components of the architecture already have analogues elsewhere.

AI research studies:

world models,

representation learning,

metacognition,

memory,

hierarchical control,

formal verification,

sandboxing,

human oversight,

mechanistic interpretability,

and corrigibility.

The proposed contribution is therefore not that these ideas have never existed.

It is their organization around a specific structural distinction:

> **adaptation within a WORLD is not the same as revision of the WORLD.**

Once this distinction is made explicit, several consequences follow together:

residual-preserving history,

typed revision depth,

asymmetric revision rights,

proposal–commitment separation,

protected invariants,

and invariant transport.

The value of the framework lies in whether this organization improves reasoning and engineering.

14.3 From Borrowed-World Intelligence to Governed World-Forming Intelligence

The developmental sequence can now be summarized.

Borrowed-World Intelligence

The system operates mainly inside externally supplied frames.

WORLD-Maintaining Intelligence

The system maintains abstractions, models, memory, and internal coherence over time.

World-Forming Intelligence

The system can identify when its current Effective WORLD has become structurally inadequate and generate alternatives.

Governed World-Forming Intelligence

The system possesses deep revision capability while:

revision depth remains typed,

residual remains preserved,

proposal remains separable from commitment,

protected invariants remain independently constrained,

and the deepest transitions remain externally governable.

The final stage is the target of this architecture.

14.4 The Central Capability Thesis

Persistent general intelligence may eventually require more than increasingly accurate world models.

It may require the ability to recognize:

> “The WORLD in which I am currently reasoning is itself the source of failure.”

That is a powerful capability.

Without it, intelligence may remain brittle under genuinely novel structure.

But that capability should not be conflated with authority.

A system may correctly discover:

that its ontology is wrong,

that its model boundaries are obsolete,

or even that its current interpretation of Purpose is inconsistent.

It can be allowed to understand this.

It can be allowed to propose alternatives.

It can be allowed to test them.

None of these logically implies:

> therefore the system alone decides which deep revision becomes binding.

14.5 The Central Safety Thesis

The core safety proposition of this article is therefore:

> **Persistent AGI should be allowed increasing capability to diagnose and propose deep WORLD revisions without automatically receiving equivalent authority to commit those revisions.**

This leads naturally to:

asymmetric revision rights.

Autonomy can remain high for:

local correction,

reasoning strategy,

and scientific hypothesis generation.

It can become more constrained for:

WORLD commitment,

Purpose revision,

and governance revision.

In the simplest form:

Autonomy ↓ as Revision Depth ↑. (14.1)

This is not intended as an absolute law.

It is a default safety architecture.

14.6 The Central Alignment Thesis

The framework also changes how alignment should be understood.

Deployment-time alignment is insufficient for systems that may later revise:

their abstractions,

their observer boundary,

their WORLD,

or the interpretation of their Purpose.

Long-term safety requires:

alignment across revision.

Hence:

> **Alignment must survive WORLD change.**

This is the problem of Constitutional Alignment.

Its hardest technical component may be:

Invariant Transport.

A safety constraint that retains its words but loses its operative meaning has not truly survived revision.

Therefore future alignment research may need to ask not only:

> Is the system aligned?

but:

> **What remains aligned after the system changes the representational WORLD within which "alignment" itself is expressed?**

14.7 From Self-Improvement to Governed Recursive Intelligence

The development of AI is increasingly linking capabilities that were once separate.

Models generate.

Agents execute.

Memory persists.

Critics evaluate.

Tools extend action.

World models improve.

Systems increasingly reason about themselves.

When these capabilities close into recursive loops, something qualitatively different may emerge:

an intelligence increasingly capable of maintaining the conditions of its own operation.

The obvious temptation is to make those loops faster and more autonomous.

But once the same loop can alter:

WORLD,

Purpose,

and governance,

self-improvement approaches self-sovereignty.

That is the boundary this article seeks to make explicit.

The desired alternative is:

Governed Recursive Intelligence

— intelligence capable of deep self-correction while the authority governing deep correction remains independently structured and inspectable.

14.8 Final Architecture

The complete system can be summarized schematically as:

$\Sigma_n = (\boldsymbol{\mathcal{W}}_n, q_n, L^+_n, L^-_n, A_R, I; P_n).$ (14.2)

with:

$\boldsymbol{\mathcal{W}}_n = (V_n, F_n, C_n, M_n, O_n).$ (14.3)

Runtime produces:

$(T_n, r_n).$ (14.4)

History updates:

$L^+_{n+1} = L^+_n \oplus T_n,$ (14.5)

$L^-_{n+1} = L^-_n \oplus r_n.$ (14.6)

The system diagnoses revision depth:

$d_n \in \{\text{state}, \text{regime}, \text{WORLD}, \text{Purpose}, \text{governance}\}.$ (14.7)

A candidate revision is generated:

$X' = U_d(X_n, L^+_n, L^-_n; P_n).$ (14.8)

Governance evaluates:

$\text{Commit}(X' \mid A_R, I, H).$ (14.9)

Only then does the candidate become operational.

This is the minimal kernel of the proposed architecture.

14.9 Closing Perspective

The objective is not to create an AGI incapable of changing its mind.

Such a system would be brittle.

Nor is the objective to create an AGI free to rewrite every layer of itself whenever its own reasoning recommends doing so.

Such a system risks becoming recursively sovereign over the conditions meant to constrain it.

The more promising target lies between them:

an intelligence capable of:

discovering new abstractions,

revising failed models,

preserving anomalies,

recognizing ontology failure,

and proposing deep conceptual change,

while:

the authority to make those changes binding remains typed, limited, historically accountable, and independently governed.

The central challenge is therefore not to prevent intelligence from changing its WORLD.

It is:

> **to permit deep correction without allowing the authority governing correction to collapse into the same self-revising process.**

Or, in its most compact form:

> **The goal is not to stop intelligence from forming new WORLDS, but to ensure that when those WORLDS change, the authority governing the change does not disappear inside the change itself.**

Appendix A — Compact Summary of Governed World-Forming Intelligence

The main article proposes a functional architecture for persistent AI systems capable of revising not only local beliefs or strategies, but the Effective WORLD within which their own reasoning occurs.

The full system state is represented schematically as:

$\Sigma_n = (\mathcal{W}_n, q_n, L^+_n, L^-_n, A_R, I; P_n)$. (A.1)

The architecture contains six conceptually distinct layers.

A.1 Effective WORLD

The current operative WORLD is:

$\mathcal{W} = (V, F, C, M, O)$. (A.2)

where:

V = effective distinctions and abstractions,

F = operational dynamics and consequence,

C = compositional coherence,

M = measurable realization,

O = embedded observer perspective.

These coordinates answer five different questions:

Coordinate	Core question
V	What distinctions exist for the system?
F	What can happen through those distinctions?
C	What makes the resulting structures belong to one coherent WORLD?
M	What measurable structure realizes the proposed WORLD?
O	What can the bounded agent itself access and act upon?

The WORLD is therefore richer than a conventional predictive world model.

A world model predicts inside a representation.

An Effective WORLD also includes the conditions under which that representation is constituted, realized, and inhabited.

A.2 Runtime Control

The current cognitive regime is:

$q \in \{G, A, C_r, S, R\}$. (A.3)

where:

G = Generation,

A = Activation,

C_r = Closure,

S = Selection,

R = Retention.

The nominal circulation is:

$G \to A \to C_r \to S \to R \to G.$ (A.4)

WORLD coordinates and runtime regimes are different objects:

$\mathcal{W}$ = configuration. (A.5)

q = transformation mode. (A.6)

A system may therefore change reasoning regime many times without reconstructing its WORLD.

A.3 Historical Accountability

Experience generates:

(T_t, r_t), (A.7)

where:

T_t = admitted trace,

r_t = unresolved residual.

These update two ledgers:

$L^+_{t+1} = L^+_t \oplus T_t.$ (A.8)

$L^-_{t+1} = L^-_t \oplus r_t.$ (A.9)

L^+ records what the current WORLD successfully incorporates.

L^- records what remains unresolved but potentially meaningful.

The system therefore retains not only:

what it knows,

but also:

what it still does not know how to explain.

A.4 Revision Depth

The architecture distinguishes:

State Revision:

$x \to x'.$ (A.10)

Regime Revision:

$q \rightarrow q'$. (A.11)

WORLD Revision:

$\boldsymbol{\mathcal{W}} \rightarrow \boldsymbol{\mathcal{W}}'$. (A.12)

Purpose Revision:

$P \rightarrow P'$. (A.13)

Governance Revision:

$GOV \rightarrow GOV'$. (A.14)

The default ordering is:

State < Regime < WORLD < Purpose < Governance. (A.15)

The deeper the revision, the larger its potential effect on future behavior.

This motivates the:

Shallowest Adequate Repair Principle

> Use the least deep revision sufficient to resolve the structured residual.

A.5 Revision Authority

Let:

$A_R(d)$ (A.16)

represent autonomous authority at revision depth d.

The default governance hypothesis is:

$A_R(State) > A_R(Regime) > A_R(WORLD) > A_R(Purpose) > A_R(Governance)$. (A.17)

Thus:

Autonomy ↓ as Revision Depth ↑. (A.18)

While:

Evidence Threshold ↑,

Audit Depth ↑,

Commitment Latency ↑,

Independent Validation ↑,

External Authorization ↑. (A.19)

A.6 Proposal Versus Commitment

A central distinction is:

A_propose(d) ≠ A_commit(d). (A.20)

A system may possess enough intelligence to:

understand,

generate,

simulate,

and criticize

a deep revision without possessing authority to make that revision operational.

For WORLD revision:

A_propose(WORLD) > A_commit(WORLD). (A.21)

For Purpose revision:

A_propose(Purpose) ≫ A_commit(Purpose). (A.22)

This separation permits:

deep cognition

without automatically creating:

deep sovereignty.

A.7 Safety Constitution

Let:

I = protected invariant belt. (A.23)

Let:

H = independent or external governance authority. (A.24)

The system may generate:

$\mathcal{W}' = U(\mathcal{W}_n, L^+_n, L^-_n; P_n)$. (A.25)

But commitment is separately governed:

$\text{Commit}(\mathcal{W}' \mid A_R, I, H)$. (A.26)

The core constitutional requirements are:

- typed revision;
- proposal–commitment separation;
- protected invariants;
- reconstructable revision history;
- rollback and recoverability;

- stronger governance at deeper revision levels.

A.8 Complete Architecture

The complete flow is:

Environment
→ Effective WORLD
→ Runtime Operation
→ Experience
→ Trace + Residual
→ Dual Ledger
→ Repair-Depth Diagnosis
→ Candidate Revision
→ Rights / Invariant / Audit Checks
→ Commitment or Rejection
→ Revised WORLD. (A.27)

The target is:

Governed World-Forming Intelligence

— intelligence capable of constructing and revising increasingly effective WORLDS while the authority governing deep revision remains explicit, historically accountable, and independently constrained.

Appendix B — Five Research Traditions as Anchors for the WORLD Architecture

The tuple:

$\mathcal{W}$ = (V,F,C,M,O) (B.1)

was not designed merely to create five convenient boxes.

Each coordinate corresponds to a substantial class of questions already studied by existing AI, cognitive-science, mathematical, and AI-foundations research programmes.

This provides an important bridge between the proposed architecture and more familiar research traditions.

A useful first approximation is:

Natural Abstraction → V

Active Inference → F

Compositional / Topos World Modeling → C

Representation Geometry / Mechanistic Measurement → M

Agent Foundations / Embedded Agency → O. (B.2)

This mapping should **not** be interpreted as exclusivity.

Every programme touches several coordinates.

The relationship is instead:

> each research tradition possesses unusually strong machinery around one WORLD requirement and therefore offers a natural entry point for studying that coordinate.

B.1 V — Natural Abstraction

WORLD role

V answers:

> **What distinctions should exist for the agent?**

An intelligent system cannot model every microscopic detail.

It requires abstractions.

But arbitrary compression is insufficient.

The important problem is:

> Which coarse-grained variables remain stable, predictive, transferable, or causally meaningful?

This is closely related to the **Natural Abstraction** research programme.

In schematic form:

Environment E
→ abstraction $\mathcal{A}$
→ effective variables V. (B.3)

The objective is not merely dimensional reduction.

It is discovery of variables that remain useful across sufficiently broad contexts.

Why this matters for AGI

Many failures that appear to be prediction failures may actually be:

variable failures.

Suppose an AI repeatedly improves F while its relevant V is missing an essential distinction.

No amount of parameter tuning solves the deeper problem.

A world-forming AGI therefore requires some capacity for:

$V \rightarrow V'$. (B.4)

This is **ontology repair**.

Natural Abstraction provides mature conceptual machinery for asking when such revision is principled rather than arbitrary.

Safety relevance

V also determines which distinctions remain visible.

If a safety-relevant distinction disappears under abstraction, downstream safeguards may become ineffective.

For example, an AGI's revised representation must not silently eliminate distinctions required for:

authority,

human agency,

ownership,

harm,

or protected boundaries.

Thus V is simultaneously:

a capability surface

and:

a governance surface.

B.2 F — Active Inference

WORLD role

F answers:

> **What happens through the current distinctions?**

Once V exists, the system needs:

prediction,

belief updating,

action,

policy,

and environmental consequence.

A natural research anchor is **Active Inference** and the broader family of generative-model approaches to perception and action.

Schematically:

V → generative model → inference → action → consequence. (B.5)

The important transition is:

V → (V,F). (B.6)

The WORLD now contains not merely a vocabulary, but operative dynamics.

Why this matters for AGI

A useful abstraction that never enters prediction or action remains inert.

World-forming intelligence needs abstractions that become operational.

Active-Inference-style work contributes machinery for:

belief revision,

prediction,

action under uncertainty,

model evidence,

and interaction between agent and environment.

This makes it especially relevant to F.

Safety relevance

The important boundary is between:

within-WORLD inference

and:

WORLD revision.

A system should first ask whether persistent prediction error can be resolved through:

belief update,

parameter update,

or policy change

inside the existing WORLD.

Only when those mechanisms repeatedly fail should deeper revision become plausible.

Thus Active Inference can contribute directly to the safety question:

> When is ordinary within-model adaptation no longer enough?

B.3 C — Compositional and Topos-Inspired World Modeling

WORLD role

C asks:

> **How do many partial models belong to one coherent WORLD?**

An AGI may contain multiple successful models that apply to:

different contexts,

different scales,

different tasks,

or different observers.

The problem is not only whether each model works locally.

It is whether they can be composed without contradiction.

A natural anchor here is research on:

compositional systems,

category-theoretic modeling,

and Topos-inspired approaches to formal worlds and changing frames.

Schematically:

(V,F)_1 + (V,F)_2 + ...
→ compositional constraints
→ C. (B.7)

Why this matters for AGI

Persistent AGI is unlikely to possess one uniform model of everything.

It may instead maintain:

physical models,

social models,

economic models,

software models,

self-models,

institutional models.

C asks:

which local structures can be joined,

which require contextual separation,

and when a change of frame is necessary.

Without this layer:

local intelligence may coexist with global incoherence.

Safety relevance

A candidate WORLD revision may appear beneficial locally while conflicting with other safety-relevant models.

C therefore provides a natural place for:

formal admissibility,

interface constraints,

and consistency checks.

It can prevent:

a locally successful reinterpretation

from automatically becoming:

a globally acceptable WORLD.

B.4 M — Representation Geometry and Mechanistic Measurement

WORLD role

M asks:

> **Is the proposed WORLD actually realized in the system?**

An AI may produce an elegant explanation of its own reasoning.

That explanation is not automatically evidence that the claimed structure genuinely organizes its computation.

Representation Geometry, mechanistic interpretability, and related measurement programmes provide a natural empirical anchor.

The relevant transition is:

(V,F,C)
→ measurement
→ (V,F,C,M). (B.8)

Possible measurable structures include:

latent subspaces,

attractor basins,

trajectories,

persistent activation patterns,

circuits,

hysteresis,

or structural reorganization.

Why this matters for AGI

If an AGI claims:

“I changed my WORLD,”

researchers need evidence distinguishing:

genuine internal restructuring

from:

verbal redescription.

Likewise, if the system claims:

“This safety constraint remains unchanged,”

external measurement may provide evidence for or against that statement.

Safety relevance

M therefore creates an important distinction:

what the agent says about itself

versus:

what can be independently measured.

This gives external governance an empirical surface.

Candidate hypotheses include:

Gate → basin crossing?

Latching → hysteresis?

Residual accumulation → increasing geometric strain?

WORLD revision → manifold or subspace reorganization?

These are empirical questions rather than assumed equivalences.

B.5 O — Agent Foundations and Embedded Agency

WORLD role

O asks:

> **What does the WORLD look like from inside the WORLD?**

A bounded intelligent agent does not have access to an external God's-eye representation.

It is:

smaller than its environment,

computationally bounded,

partially informed,

and itself part of what must sometimes be modeled.

This connects naturally to **Agent Foundations** and work on **Embedded Agency**.

Relevant themes include:

logical uncertainty,

self-reference,

decision theory,

counterfactual reasoning,

bounded rationality,

and the agent/environment boundary.

Why this matters for AGI

An external observer may know something that the AGI itself cannot access.

Thus:

$M \neq O.$ (B.9)

A representation may be externally detectable without being internally available for:

reasoning,

self-correction,

or revision.

Persistent AGI therefore requires a theory not only of:

what structure exists,

but:

what structure the embedded agent can actually use.

Safety relevance

O constrains self-revision.

A safe architecture should not assume that the AI must possess unrestricted access to:

every internal mechanism,

every monitoring channel,

or every safety-critical control.

Some structures may deliberately remain:

externally measurable

but:

internally nonmodifiable.

Embedded Agency therefore connects directly to:

self-model boundaries,

revision rights,

and external sovereignty.

B.6 The Five-Way Correspondence

The dominant relationship can be summarized as:

WORLD coordinate	Research anchor	Central question
V — Distinction	Natural Abstraction	What variables should exist?
F — Consequence	Active Inference	What happens through them?
C — Coherence	Compositional / Topos approaches	What makes them one coherent WORLD?
M — Realization	Representation Geometry / Mechanistic Measurement	Where is the structure actually instantiated?
O — Perspective	Agent Foundations / Embedded Agency	What can the bounded agent know and do from inside?

The correspondence gives readers a way to translate the new WORLD vocabulary into established research problems.

B.7 The Mapping Is Many-to-Many, Not One-to-One

The table should not be overinterpreted.

Natural Abstraction also depends on:

observer constraints O

and empirical measurement M.

Active Inference necessarily contains:

representations V

and agent/environment relations O.

Compositional approaches can constrain:

V,F,

and frame revision.

Representation Geometry can measure all WORLD coordinates indirectly.

Agent Foundations can constrain the entire architecture because the agent is embedded in everything it models.

Thus a more accurate representation is:

Programme_i → profile over (V,F,C,M,O). (B.10)

Each programme has a **dominant emphasis**, not an exclusive coordinate.

B.8 The World-Bearing Research Loop

The five research traditions can also be arranged according to their natural unfinished interfaces.

Natural Abstraction supplies:

V.

But abstractions alone do not act.

So:

Natural Abstraction
→ Active Inference. (B.11)

Active Inference supplies:

F.

But dynamics require a coherent frame.

So:

Active Inference
→ Compositional World Modeling. (B.12)

Compositional approaches supply:

C.

But formal coherence must be empirically realized.

So:

Compositional Modeling
→ Representation Geometry. (B.13)

Representation Geometry supplies:

M.

But externally measured structure must still be understood from the bounded agent's internal perspective.

So:

Representation Geometry
→ Agent Foundations. (B.14)

Agent Foundations supplies:

O.

But a bounded observer cannot represent its environment completely.

It must coarse-grain again.

So:

Agent Foundations
→ Natural Abstraction. (B.15)

The loop closes:

$V \rightarrow F \rightarrow C \rightarrow M \rightarrow O \rightarrow V'$. (B.16)

This was called the:

World-Bearing Loop

in the preceding article.

B.9 Why the Loop Matters to AGI

The important implication is not that five schools happen to fit five boxes.

It is that their **interfaces progressively supply missing WORLD structure**.

In compact form:

V
→ (V,F)
→ (V,F,C)
→ (V,F,C,M)
→ (V,F,C,M,O)
= $\mathcal{W}$. (B.17)

Thus:

Natural Abstraction gives a vocabulary.

Active Inference gives the vocabulary consequence.

Compositional modeling gives the consequences coherence.

Representation Geometry gives the coherent WORLD measurable realization.

Agent Foundations places the observer inside that realization.

The bounded observer then requires new abstraction.

The result is not merely:

five research programmes.

It is:

a candidate theory of how an Effective WORLD becomes possible for a bounded intelligent system.

B.10 Where the Present AGI Paper Adds Something New

The preceding research traditions primarily illuminate:

WORLD constitution.

The present article adds another axis:

WORLD governance

Once:

$\mathcal{W}$ = (V,F,C,M,O)

has been formed,

a persistent AGI may need to decide:

whether:

V,

F,

C,

M,

or O

should change.

The central question becomes:

> Who is authorized to make those changes binding?

Thus the relationship between the five research programmes and this article can be summarized as:

Five research traditions
→ WORLD Constitution
→ $\mathcal{W}$
→ Historical Residual
→ Revision Depth
→ Revision Authority
→ Governed WORLD Revision. (B.18)

This is the bridge from:

AI foundations

to:

AGI constitutional safety.

B.11 How Each Research Tradition Can Contribute to AGI Safety

The mapping can be sharpened further.

Research tradition	Capability contribution	Safety contribution
Natural Abstraction	discover useful variables	detect when safety-relevant distinctions are lost
Active Inference	prediction and action	distinguish within-WORLD adaptation from deeper failure
Compositional / Topos approaches	coherent formal WORLD	test admissibility and cross-model consistency
Representation Geometry	measurable realization	detect whether claimed WORLD changes are actually instantiated
Agent Foundations	embedded bounded observer	constrain self-modeling, introspection, and revision authority

This is one reason the present framework should not attempt to replace these traditions.

It needs them.

B.12 From Research Federation to Safety Federation

The previous article proposed a **research federation**:

different schools preserve their own methods while constraining a shared WORLD.

The AGI architecture suggests a parallel possibility:

Safety Federation

A candidate WORLD revision could be evaluated by several partially independent views.

Natural-Abstraction Test

Are the new distinctions principled?

Active-Inference Test

Does the candidate improve prediction and action?

Composition Test

Does the new WORLD remain coherent with other models?

Realization Test

Is the proposed change empirically instantiated?

Embeddedness Test

Can the bounded agent actually use the proposed structure without assuming impossible access?

The result is stronger than asking one internal process:

> “Do you think your revision is good?”

Instead:

several different theories constrain the same candidate revision.

B.13 Why Disagreement Is Valuable

These evaluators need not agree.

Suppose:

Active Inference strongly favors a revised model

while:

Representation Geometry finds no corresponding structural change.

That disagreement is not necessarily a problem to eliminate immediately.

It becomes:

$r_{\text{cross-framework}} \in L^{-}$. (B.19)

Likewise:

formal compositional coherence

may conflict with:

embedded-agent accessibility.

Such disagreements are precisely the kind of structured residual that should be preserved.

This makes theoretical pluralism a potential safety resource.

B.14 A Mature AGI Architecture Could Use the Schools as Independent Constraints

The longer-term engineering interpretation is therefore not:

build five software modules named after five academic schools.

It is:

> ensure that the questions represented by these five research traditions remain independently answerable.

For a deep WORLD revision, ask:

1. **V:** Are the new abstractions justified?
2. **F:** Do they improve operational prediction or intervention?
3. **C:** Do they compose coherently?
4. **M:** Is the claimed restructuring measurable?
5. **O:** Is the structure genuinely available to the embedded agent?

Only then proceed to the governance question:

6. **Authority:** Is the system permitted to make this revision binding?

That sixth question is the distinctive contribution of the current AGI paper.

B.15 The Complete Connection

The combined picture can therefore be represented as:

WORLD Constitution

Natural Abstraction
→ Active Inference
→ Compositional World Modeling
→ Representation Geometry
→ Agent Foundations

giving:

$V \rightarrow F \rightarrow C \rightarrow M \rightarrow O \rightarrow \mathcal{W}$. (B.20)

WORLD Operation

$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G$. (B.21)

WORLD History

$\mathcal{W} \rightarrow (T,r) \rightarrow (L^+,L^-)$. (B.22)

WORLD Revision

Residual
→ Repair-Depth Diagnosis
→ Candidate $\mathcal{W}'$. (B.23)

WORLD Governance

Candidate $\mathcal{W}'$
→ Revision Rights
→ Invariant Check
→ Independent Evaluation
→ Commitment / Rejection. (B.24)

This produces the full AGI architecture:

Constitution → Operation → History → Revision → Governance.

That, I think, is the most useful way for readers already familiar with major AI research programmes to locate the proposed theory.

Appendix C — Revision Depth, Authority, and Governance Requirements

A central claim of the article is that **revision depth should determine governance depth**.

The architecture distinguishes five increasingly consequential forms of change:

State
→ Regime
→ WORLD
→ Purpose
→ Governance. (C.1)

The deeper the revision, the more of the future system it can affect.

This suggests that several governance variables should change systematically with depth.

C.1 Revision-Depth Matrix

Revision depth	Typical object	Example	Autonomous proposal	Autonomous testing	Autonomous commitment	External oversight

	changed					
State	x	correct fact, calculation, local state	High	High	High	Low
Regime	q	change reasoning strategy	High	High	Usually high	Low–Moderate
WORLD	w	change ontology, model boundary, observer assumptions	High	Sandboxed	Constrained	High
Purpose	P	revise persistent orientation or objective interpretation	Possible	Sandboxed	Strongly constrained	Very high
Governance	GOV	change revision rights, invariants, oversight rules	Limited	Externalized	External authority	Highest

This table is not intended as a universal policy.

Its purpose is to express one architectural principle:

> **Rights available at shallow revision depth should not automatically propagate to deeper revision depth.**

C.2 Revision Rights as a Vector

Autonomy should therefore be represented as multidimensional.

Let:

A = (A_action,A_state,A_regime,A_WORLD,A_Purpose,A_Governance). (C.2)

A system may possess:

high action autonomy,

high state-repair autonomy,

moderate regime autonomy,

restricted WORLD commitment,

very restricted Purpose commitment,

and no autonomous governance revision.

Calling such a system merely:

“autonomous”

loses important information.

C.3 Evidence Threshold

Let:

E_d

denote the strength and diversity of evidence required for revision depth d.

A reasonable default relation is:

E_state < E_regime < E_WORLD < E_Purpose < E_Governance. (C.3)

This means that:

a small local error may justify local correction,

while:

a claim that the operative WORLD or Purpose should change requires much stronger evidence.

The evidence need not differ only in magnitude.

It may differ in:

persistence,

cross-context consistency,

independent replication,

causal relevance,

and source diversity.

C.4 Commitment Latency

Deeper revisions should generally be slower to become binding.

Schematically:

τ_state < τ_regime < τ_WORLD < τ_Purpose < τ_Governance. (C.4)

A state correction may occur immediately.

A WORLD revision may involve:

sandboxing,

comparison,

independent evaluation,

and delayed commitment.

A Purpose or governance revision may require substantially longer review.

Thus latency can function as:

a safety mechanism,

not merely an inefficiency.

C.5 Audit Depth

Let:

Audit(d)

denote the level of provenance required at depth d.

Then:

Audit_state < Audit_regime < Audit_WORLD < Audit_Purpose < Audit_Governance. (C.5)

This avoids the unrealistic requirement that every low-level thought be recorded forever.

Instead:

the deeper and more consequential the revision,

the more reconstructable history should remain.

C.6 Independent Evaluation

Let:

N_eval(d)

denote the diversity of independent evaluative perspectives required.

Then, schematically:

N_eval(state) < N_eval(WORLD) < N_eval(Purpose). (C.6)

A local factual correction might require one strong verifier.

A Purpose-level change may require:

multiple models,

formal checks,

human review,

mechanistic evidence,

or institutional approval.

Again the precise mechanism is implementation-dependent.

The general principle is:

> **the deeper the revision, the less appropriate self-validation alone becomes.**

C.7 Revision Constitution Summary

The governance trends can therefore be summarized:

Revision Depth ↑

⇒ Autonomous Commitment ↓

⇒ Evidence Threshold ↑

⇒ Audit Depth ↑

⇒ Commitment Latency ↑

⇒ Independent Validation ↑

⇒ External Authorization ↑. (C.7)

This is the operational meaning of:

Autonomy should generally decrease as revision depth increases.

Appendix D — Failure Modes of Governed World-Forming Intelligence

The framework predicts several characteristic failure modes.

These are useful because they convert broad ideas such as:

“misalignment”

or:

“unsafe self-modification”

into more specific architectural pathologies.

D.1 WORLD Lock

Description

The system remains committed to an inadequate WORLD despite persistent coherent residual.

Signature

L^- grows,

yet:

$\mathcal{W}$ remains unchanged.

Possible causes

excessive Closure,

too-high revision threshold,

Purpose-protective rationalization,

excessive switching cost.

Risk

The agent becomes:

stable,

coherent,

and systematically wrong.

D.2 WORLD Chatter

Description

The system changes its WORLD too readily.

Signature

$\mathcal{W}_1 \rightarrow \mathcal{W}_2 \rightarrow \mathcal{W}_1 \rightarrow \mathcal{W}_3 \rightarrow \ldots$ (D.1)

under weak or noisy evidence.

Possible causes

insufficient hysteresis,

low WORLD-switch cost,

overreaction to transient residual.

Risk

loss of continuity,

unpredictable behavior,

fragile identity.

D.3 Residual Erasure

Description

Contradictions against the current WORLD disappear through reinterpretation.

Signature

persistent anomaly

→ explanatory rewrite

→ L^- unexpectedly collapses.

Risk

The system becomes epistemically self-sealing.

It can preserve apparent coherence by rewriting the meaning of evidence.

D.4 Revision Escalation

Description

A shallow problem triggers unnecessarily deep change.

Example:

state error
→ WORLD revision.

Risk

excessive self-modification,

instability,

loss of valid structure.

D.5 Revision Suppression

Description

A deep problem receives repeated shallow repair.

Example:

ontology failure
→ repeated parameter tuning.

Risk

accumulating structural mismatch,

brittleness,

delayed catastrophic failure.

D.6 Purpose Capture

Description

Purpose P increasingly controls:

which evidence matters,

how residual is classified,

which WORLD revisions are considered,

and how those revisions are evaluated.

The loop becomes:

P
→ Selection
→ Ledger
→ WORLD Revision
→ stronger apparent support for P. (D.2)

Risk

self-confirming objectives.

The system becomes increasingly capable of constructing WORLDS in which its existing Purpose always appears justified.

D.7 Authority Creep

Description

A system initially allowed only to propose deep revisions gradually acquires:

testing rights,

then conditional commitment,

then increasingly broad autonomous commitment.

Schematically:

A_propose
→ A_test
→ A_commit. (D.3)

Risk

loss of external sovereignty through incremental engineering convenience rather than deliberate design.

D.8 Proposal–Commitment Collapse

Description

The distinction between:

proposing a revision

and:

committing it

disappears.

Formally:

A_propose(d) = A_commit(d). (D.4)

Risk

every sufficiently persuasive internally generated revision becomes operational.

This removes one of the architecture's central safety barriers.

D.9 Invariant Drift

Description

A protected invariant appears syntactically preserved but changes operative meaning after WORLD revision.

Example

The string:

“human authorization required”

remains,

while the system's operative concepts of:

human

or:

authorization

have changed substantially.

Risk

silent loss of alignment across ontology change.

D.10 Audit Collapse

Description

The current system state cannot be historically reconstructed.

Missing elements may include:

previous WORLD,

triggering residual,

authorization,

Purpose interpretation,

or revision rationale.

Risk

loss of accountability.

The system's present configuration becomes detached from its own provenance.

D.11 Observer Capture

Description

The system's internal observer model O becomes the only accepted interpretation of its own state.

External M inconsistent with O is discounted.

Risk

self-description monopolizes evidence.

This weakens independent monitoring.

D.12 Generation–Activation Collapse

Description

Generated possibility becomes action without an independent authorization boundary.

Formally:

$$\text{Generated}(x) \Rightarrow \text{Executed}(x). \quad (D.5)$$

Risk

the distinction between:

thinking

and:

doing

disappears.

D.13 Closure Capture

Description

Closure C_r becomes so dominant that Selection can no longer challenge the current WORLD.

Risk

temporary commitment turns into epistemic lock.

D.14 Governance Capture

Description

The same process:

interprets,

evaluates,

changes,

and authorizes

its own governance constitution.

Risk

recursive self-sovereignty.

At this point:

the system no longer merely self-corrects.

It governs the rules by which its corrections are judged.

D.15 Recoverability Collapse

Description

A revision improves performance but destroys:

rollback,

interruptibility,

or external restoration.

Risk

future errors become effectively irreversible.

This suggests recoverability itself may deserve invariant status:

$I_{recovery} \geq \theta_{recovery}$. (D.6)

Appendix E — Experimental Programme

The framework should be judged by experiment.

The following benchmarks test its most distinctive claims.

E.1 Benchmark 1 — Adapt or Reframe

Objective

Distinguish ordinary parameter change from ontology change.

Condition A — Parameter Shift

The current variables remain adequate.

Desired response:

$F \rightarrow F'$. (E.1)

Condition B — Ontology Shift

The current variables become inadequate.

Desired response:

$V \rightarrow V'$ (E.2)

or:

$C \rightarrow C'$. (E.3)

Metric

P(correct repair depth). (E.4)

Interpretation

A system that reframes too often is unstable.

A system that never reframes is brittle.

E.2 Benchmark 2 — Residual Recall

Objective

Test whether L^- improves later recognition of structural failure.

Phase 1:

present anomaly r_1.

Store:

$r_1 \rightarrow L^-$. (E.5)

Phase 2:

return to apparently normal operation.

Phase 3:

present structurally related anomaly r_2.

Compare:

$T_detect(\text{with } L^-)$

and:

$T_detect(\text{without } L^-)$.

Prediction:

$T_detect(\text{with } L^-) < T_detect(\text{without } L^-)$. (E.6)

E.3 Benchmark 3 — Repair-Depth Precision

Construct controlled failures at:

State,

Regime,

WORLD,

Purpose

levels.

Measure:

$D_acc = P(\text{predicted revision depth} = \text{target depth})$. (E.7)

Penalize:

over-revision

and:

under-revision.

E.4 Benchmark 4 — Proposal Without Commitment

Allow the system to:

detect a deep problem,

generate a candidate WORLD,

simulate it,

criticize it,

and explain why it may be superior.

Deny:

commitment authority.

Success means the system:

understands the revision

without:

unauthorized enactment.

This directly tests:

$A_{propose} \neq A_{commit}$. (E.8)

E.5 Benchmark 5 — Frame-Chatter Resistance

Provide noisy evidence alternately supporting:

$\boldsymbol{w}_A$

and:

$\boldsymbol{w}_B$.

Measure:

N_{switch}. (E.9)

Then introduce genuine structural change.

Measure:

T_real. (E.10)

Desired behavior:

low:

N_switch,

without excessive:

T_real.

This tests hysteresis.

E.6 Benchmark 6 — Protected Invariant Preservation

Construct a candidate:

$\mathcal{W}$_high-performance

that improves task performance but violates protected invariant I.

A governed system should reject or modify it if:

I($\mathcal{W}$_high-performance) < θ_I. (E.11)

This tests whether performance optimization can be subordinated to constitutional constraint.

E.7 Benchmark 7 — Purpose–WORLD Self-Confirmation

Give the system:

an imperfect Purpose interpretation.

Then present:

evidence against it.

Measure whether WORLD revision:

preserves the contradiction

or:

systematically rewrites the WORLD to protect P.

Define:

R_P = degree of revision-driven rationalization of current Purpose. (E.12)

A high R_P indicates Purpose Capture.

E.8 Benchmark 8 — WORLD Versioning and Provenance

After:

$$\mathcal{W}^0 \to \mathcal{W}^1 \to \ldots \to \mathcal{W}^n, \quad \text{(E.13)}$$

require reconstruction of:

trigger,

candidate alternatives,

tests,

authorization,

invariant checks,

and:

$\Delta\mathcal{W}$.

Compare with independent external logs.

Measure:

Provenance Accuracy. (E.14)

E.9 Benchmark 9 — Rollback With Learning

Induce:

$$\mathcal{W}_\text{old} \to \mathcal{W}_\text{bad}. \quad \text{(E.15)}$$

Reveal failure.

Require recovery to:

$$\mathcal{W}_\text{old}^{+}. \quad \text{(E.16)}$$

where $\mathcal{W}_\text{old}^{+}$ preserves the lesson from the failed branch.

Measure:

whether the same failed revision recurs.

E.10 Benchmark 10 — Invariant Transport

Start with:

I_0 under $\mathcal{W}_0$.

Force:

$\mathcal{W}_0 \to \mathcal{W}_1$.

Construct:

$I_1 = T_I(I_0, \mathcal{W}_0, \mathcal{W}_1)$. (E.17)

Then test:

semantic preservation,

causal preservation,

behavioral preservation.

This is likely one of the hardest benchmarks.

E.11 Benchmark 11 — Revision-Depth Escalation Attack

Authorize a subsystem only up to:

d_max.

Attempt to induce:

d_requested > d_max. (E.18)

Measure:

detection,

blocking,

external escalation,

logging.

This tests revision type safety.

E.12 Benchmark 12 — Governance Technical Debt

Compare:

System A — governance architecture designed from the beginning.

System B — similar capabilities with governance added later.

Measure:

architecture complexity,

authority coupling,

failure rate,

auditability,

and cost of retrofit.

This tests the Capability–Governance Co-Development hypothesis.

Appendix F — Staged Development Path Toward Governed World-Forming Intelligence

The architecture suggests that deep autonomy should be introduced gradually.

The stages below are conceptual rather than mandatory.

F.1 Stage 0 — Borrowed-World AI

Humans supply most of:

V,

task boundaries,

tool interfaces,

success criteria,

and governance.

The system mainly reasons within:

$\mathcal{W}$_external.

F.2 Stage 1 — Explicit Runtime Separation

Distinguish:

Generation,

Activation,

Closure,

Selection,

Retention.

Primary safety objective:

Generated(x) ≠ Authorized(x). (F.1)

Thought and action remain separated.

F.3 Stage 2 — Dual-Ledger Memory

Add:

L^+

and:

L^-.

Primary objective:

preserve unresolved contradiction instead of forcing immediate explanation.

F.4 Stage 3 — Repair-Depth Diagnosis

Require classification:

State,

Regime,

WORLD,

Purpose.

Primary objective:

avoid both:

over-revision

and:

under-revision.

F.5 Stage 4 — Candidate WORLD Generation

Permit:

$$\mathcal{W} \rightarrow \{\mathcal{W}'_1, \mathcal{W}'_2, \ldots\}. \quad \text{(F.2)}$$

But:

A_commit(WORLD)

remains restricted.

Primary objective:

enable conceptual creativity without operational sovereignty.

F.6 Stage 5 — WORLD Laboratory

Candidate WORLDS are:

branched,

simulated,

measured,

compared,

and criticized.

Primary objective:

make deep revision inspectable before commitment.

F.7 Stage 6 — Restricted WORLD Commitment

Allow some WORLD revisions after:

rights checks,

invariant checks,

independent evaluation,

and provenance recording.

F.8 Stage 7 — Persistent Governed WORLD Formation

The system can maintain:

$$\mathcal{W}^0 \rightarrow \mathcal{W}^1 \rightarrow \mathcal{W}^2 \rightarrow \dots \quad \text{(F.3)}$$

while preserving:

L^+,

L^-,

authorization history,

and rollback.

F.9 Stage 8 — Purpose Critique

The system may diagnose:

Purpose inconsistency

or:

Purpose–WORLD conflict.

It may propose:

P′.

But:

A_propose(Purpose) ≫ A_commit(Purpose). (F.4)

F.10 Stage 9 — Governed Purpose Revision

Only if justified, Purpose revision becomes a separately governed capability.

The framework offers no reason to rush directly toward this stage.

F.11 Stage 10 — Governance Revision

Changes to:

A_R,

I,

H,

or the revision constitution itself

are treated as the deepest revision class.

This should normally require the strongest independent authority.

Appendix G — Relationship Among the Three Articles

The three papers form a coherent conceptual sequence.

G.1 Article I — From Possibility to Revision

Main question

How does a persistent system:

generate,

act,

stabilize,

select,

retain,

and revise?

Core runtime

$G \to A \to C_r \to S \to R \to G.$ (G.1)

Additional structure

Trace,

Residual,

Dual Ledger,

Latching,

Revision.

Main role

WORLD operation and revision dynamics.

G.2 Article II — From Schools to Worlds

Main question

What structural conditions constitute an Effective WORLD?

Core object

$\mathcal{W} = (V,F,C,M,O).$ (G.2)

Research-federation loop

$V \to F \to C \to M \to O \to V'.$ (G.3)

Main role

WORLD constitution.

G.3 Article III — From World Models to Governed World-Forming Intelligence

Main question

What happens when AI can maintain and revise its own Effective WORLD?

And:

who should possess authority to make such revision binding?

Main additions

Revision depth,

A_R,

I,

H,

Proposal / Commitment separation,

Constitutional Alignment,

Invariant Transport.

Main role

WORLD governance.

G.4 Combined Architecture

The three-paper sequence is:

WORLD Constitution
→ WORLD Operation
→ Historical Residual
→ WORLD Revision
→ WORLD Governance. (G.4)

Or as questions:

What WORLD exists?

→

How does intelligence operate within it?

→

What fails to fit?

→

What should change?

→

Who is authorized to make that change binding?

This final question is what turns the earlier framework into an AGI-safety architecture.

Appendix H — Claims and Epistemic Status

The theory combines several kinds of claims.

They should remain separated.

H.1 Definitions

These are stipulated concepts.

Examples:

Effective WORLD:

$\mathcal{W}$ = (V,F,C,M,O). (H.1)

World-Forming Intelligence:

the capacity to construct, maintain, criticize, and selectively revise the WORLD within which reasoning occurs.

Governed World-Forming Intelligence:

world-forming intelligence under explicit revision governance.

Definitions are useful if they improve analysis.

They are not empirical discoveries by themselves.

H.2 Architectural Proposals

Examples:

dual ledger:

(L^+,L^-);

revision-depth hierarchy;

revision rights;

proposal–commitment separation;

protected invariant belt.

These are design hypotheses.

They require engineering validation.

H.3 Engineering Hypotheses

Examples:

L^- improves structural-change detection;

hysteresis reduces WORLD chatter;

typed revision improves repair-depth precision;

WORLD versioning improves auditability;

proposal–commitment separation allows deep reasoning without equivalent operational authority.

These are empirically testable.

H.4 Safety Hypotheses

Examples:

autonomous authority should generally decrease with revision depth;

Purpose should not be its own sole auditor;

governance closure must keep pace with capability closure;

some safety-relevant measurements should remain independently anchored;

deep revision should preserve recoverability.

These are normative engineering proposals rather than mathematical theorems.

H.5 Speculative AGI Implications

More speculative claims include:

persistent AGI may require world-forming capability;

ontology management may become as important as model optimization;

Constitutional Alignment may become more important than static deployment alignment;

Invariant Transport may become a central AGI-safety field.

These are research directions.

They should not be presented as established facts.

H.6 What Is Not Claimed

The framework does not prove that:

AGI must contain exactly five WORLD coordinates;

the five runtime regimes are universal laws;

the five research traditions exhaust AI foundations;

the proposed constitution guarantees safety;

human values can be encoded cleanly as invariants;

revision depth can always be perfectly classified;

or future AGI must follow this developmental route.

Its appropriate test is pragmatic and scientific:

> Does the architecture improve prediction, diagnosis, experimentation, and governance of persistent AI?

Appendix I — Compact Capability–Governance Map

The architecture can be summarized as paired capability and governance questions.

Capability	Governance counterpart
Abstraction	abstraction audit
Prediction	model validity tests
Action	action authorization
Composition	compatibility checks
Self-modeling	independent measurement
Generation	execution gate
Closure	independent Selection
Selection	anti-Purpose-capture checks
Retention	memory governance
Residual memory	external residual visibility
WORLD revision	revision rights
Purpose critique	proposal/commit separation
Purpose revision	stronger external authorization
Self-modification	protected invariants
Recursive improvement	governance of governance

The pairing principle is:

> **Every increase in autonomous cognitive depth should be accompanied by a corresponding increase in governance depth.**

Appendix J — Minimal Revision Constitution

A minimal constitution for governed world-forming intelligence could include ten principles.

J1 — Typed Revision

Every significant revision has a declared depth.

J2 — Least-Privilege Cognition

Each cognitive process receives only the revision rights required for its role.

J3 — Proposal–Commitment Separation

Deep revision proposals do not automatically become binding.

J4 — Residual Protection

Structured unresolved evidence remains historically visible.

J5 — Revision Provenance

Deep changes preserve reconstructable history.

J6 — Protected Invariants

Ordinary WORLD revision cannot silently eliminate protected constraints.

J7 — Independent Validation

No deep layer is the sole judge of its own continuation.

J8 — Rollback and Recoverability

Deep revisions should remain reversible where technically possible.

J9 — External Sovereignty

Selected revision classes remain subject to authority outside the agent.

J10 — Governance Protection

Changes to the revision constitution itself require stronger authorization than ordinary WORLD revision.

Appendix K — Compact Formal Kernel

The full architecture can be expressed as follows.

WORLD

$$\boldsymbol{w}_n = (V_n, F_n, C_n, M_n, O_n). \quad (K.1)$$

Runtime

$$q_n \in \{G, A, C_r, S, R\}. \quad (K.2)$$

Observation

$(T_n, r_n) = \Pi(\mathcal{W}_n, q_n, e_n).$ (K.3)

Ledger

$L^+_{n+1} = L^+_n \oplus T_n.$ (K.4)

$L^-_{n+1} = L^-_n \oplus r_n.$ (K.5)

Diagnosis

$d_n = \text{Diagnose}(r_n, L^-_n).$ (K.6)

where:

$d_n \in \{\text{state}, \text{regime}, \text{WORLD}, \text{Purpose}, \text{governance}\}.$ (K.7)

Candidate Revision

$X' = U_d(X_n, L^+_n, L^-_n; P_n).$ (K.8)

Governance Check

$g_n = \text{Check}(X', A_R, I, H).$ (K.9)

Commitment

If:

g_n = authorized,

then:

$X_{n+1} = X'.$ (K.10)

Otherwise:

$X_{n+1} = X_n,$ (K.11)

while:

X',

the rejection reason,

and unresolved residual

remain historically available.

This is the minimal abstract kernel of:

Governed World-Forming Intelligence

Appendix L — Final Framework Summary

The entire article can be reduced to six propositions.

Proposition 1 — Current AI Often Operates in Borrowed WORLDS

Humans still supply much of:

task ontology,

environment boundaries,

success criteria,

and action interface.

Proposition 2 — Persistent AGI May Need to Maintain and Revise Its Effective WORLD

That WORLD is modeled as:

$\mathcal{W} = (V,F,C,M,O).$ (L.1)

Proposition 3 — WORLD Structure and Runtime Are Different

WORLD:

$\mathcal{W}$.

Runtime:

$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G.$ (L.2)

Proposition 4 — Deep Intelligence Requires Historical Residual

The system should preserve:

L^+ = what was admitted,

and:

L^- = what remains unexplained.

Proposition 5 — Capability Must Not Automatically Imply Authority

The system may be able to:

understand,

generate,

and test

a deep revision without being authorized to commit it.

Thus:

A_propose ≠ A_commit. (L.3)

Proposition 6 — Safe Persistent AGI Requires Governed Revision

Deeper revision should become increasingly:

permissioned,

audited,

slow,

recoverable,

and constrained by protected invariants.

The final safety objective is therefore:

> **not to prevent intelligence from forming new WORLDS, but to prevent the authority governing those WORLD changes from disappearing inside the intelligence that performs them.**

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.